Day 21

Session 45 - Hypothesis Testing _ Part 1

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Whether the data at hand sufficiently supports a particular hypothesis.

1. Null Hypothesis (H_0) → No significant effect or relationship between variables being studied. Starting pt. for HT & represents the $\boxed{\text{status quo}}$ (no effect until proven otherwise)

2. Alternative hypothesis (H_1 or H_a)
   Statement that contradicts the null hypothesis & claims there is a significant effect or relationship between the variables being studied.

$$\boxed{\begin{aligned} H_0 &: \mu = 6\text{min} \\ H_1 &: \mu > 6\text{ min} \end{aligned}}$$

$$\boxed{\begin{aligned} H_0 &: \mu = 100\text{gm} \\ H_a &= \mu \neq 100\text{gm} \end{aligned}}$$

It represents the $\boxed{\text{research hypothesis}}$.

Imp. pts :-
i) H_0 or H_1?
ii) Evidence to reject H_0
iii) H_0 rejection failed? not means H_0 is true. It doesn't mean enough evidence to support H_1

Steps Involved

$\boxed{\text{Rejection Region Approach}}$

1. H_0 : $\mu = 6$ min
   H_a : $\mu > 6$ mins
2. Select a significance level : 0.05 or 0.01
   5% 1%
   P(rejecting H_0) when its true
3. Check assumptions: Distribution, Std, Categorical/Numerical.
4. Decide appropriate test (Z-Test, Chi-Square, ANOVA)
5. State the relevant test statistic
6. Conduct the test
7. Reject or not reject H_0
8. Interpret the results.

Z-Test E.g.1

$\mu = 50$, $\sigma = 5$

After training program, $n = 30$, $\bar{x} = 53$

Step 1) $H_0: \mu = 50$

$H_a: \mu = 53$ or $\mu > 50$

Step 2) $\alpha = 0.05$

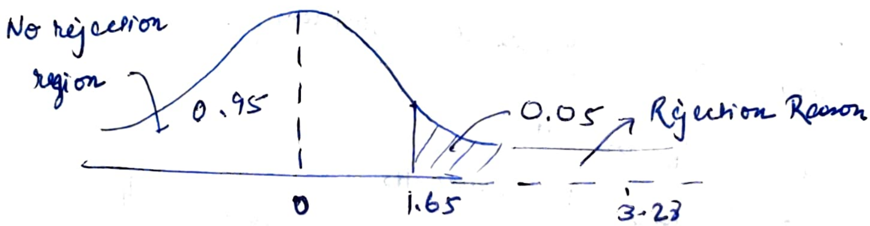
Step 3) Normality valid / σ known

4) Z-Test

5) $Z = \dfrac{\bar{x} - \mu}{\sigma/\sqrt{n}} = \dfrac{53 - 50}{5/\sqrt{30}} = \boxed{3.23}$

Rejection Region

6.



∴ We can reject the null hypothesis

7) Rejected.

E.g.2

1) $H_0: \mu = 50$

$H_a: \mu \neq 50$ (2 sided)

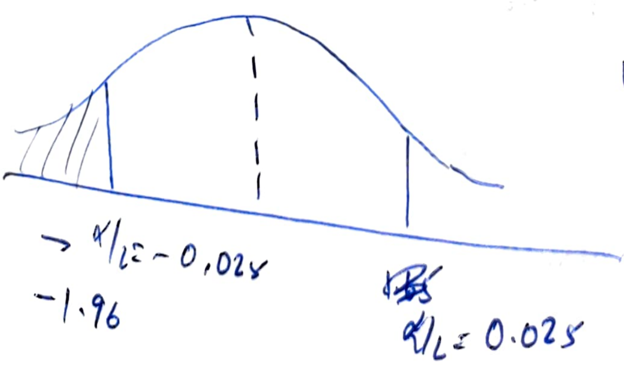
2) $\bar{x} = 49$, $\sigma = 4$ gms, $n = 40$

$\alpha = 0.05$

3) Normality valid

4) Z-Test

5) $Z = \dfrac{49 - 50}{4/\sqrt{40}} = \dfrac{-1}{\cancel{4}/\cancel{2}\, 2\sqrt{10}} = \dfrac{-\sqrt{10}}{2} = -1.581$

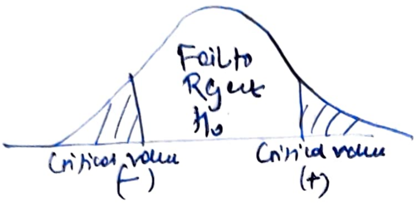


We can't reject null hypothesis

Rejection Region

Significance level → α, predetermined threshold used in hypothesis testing to determine whether H_0 should be rejected or not.

P(Rejecting the null hypothesis when its True) — Type I error



~~p-value~~ Evidence's strength is not taken into consideration

p-value Approach

α – Type-1 | β – Type-2

Type-1 vs Type-2 Error :

Truth about population

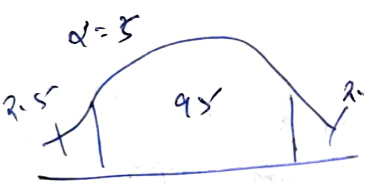
	H_0 true	Ha true (H_0 False)
Reject H_0	Type I error	✓
Accept H_0	✓	Type II error

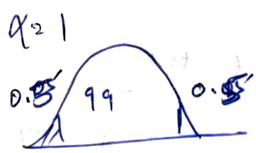
Type-1 (False positive) - α. By choosing a 'α', researchers can control risk of making Type-1 error

Type-2 (False negative)

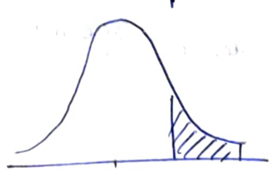
when H_0 is not rejected when it is actually.

$P(\text{Type } 2) = \beta$. Power of test = $1-\beta$



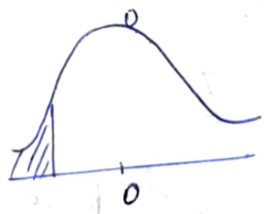


One sided vs two sided test



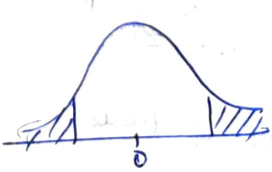
Right-tail test

Ha: μ > value



Left-tail test

Ha: μ < value



Two-Tail test

: Ha : $\mu \neq$ value

One-tailed - "$>$" or "$<$"

Advantages	Disadvantages
1. More powerful	1. Missed effects
2. Directional hypothesis	2. Increased risk of Type-I error

Two-tailed - "$\neq$"

Advantages	Disadvantages
1. Detects effects in both directions	1. Less powerful as $\alpha \to \alpha/2$ - Larger effect size or α size
2. More conservative	2. Not appropriate for directional hypothesis

Applications of Hypothesis Testing :

1. Testing effectiveness of interventions or treatments.
2. Comparing means or proportions.
3. Analysing relationships between variables :- Correlation analysis.
4. Evaluation of goodness of fit
5. Testing the independence of categorical variables
6. A/B Testing : In marketing / product development & website design

ML Applications

1. Model Comparision : Paired t-test. To compare accuracy or error rate of two models on multiple cross validation folds.
2. Feature Selection : t-test, chi-squared test, Anova
3. Hyperparameter tuning : If one set of hyperparameters lead to better per...
4. Assessing model assumptions : Can help asses whether these assumptions are met

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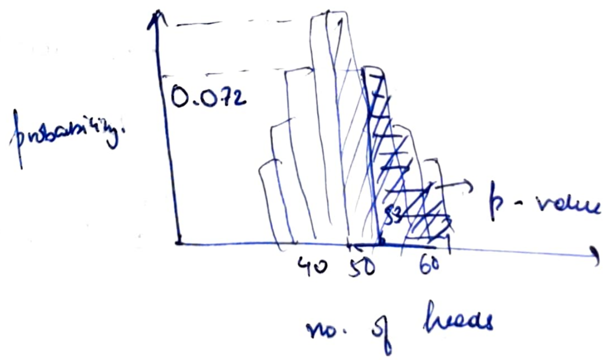
p-values & t-tests

P-value is the probability of seeing a sample as or more extreme than our own sample given having more evidence against H_0 the H_0 is true.



$H_0: P(H) = P(T)$

$H_a: P(H) > P(T)$

exp → 100 times

$P(53) = 0.0666$

$P\text{-value} = 0.3086$

$\boxed{p = 0.3} \rightarrow$ 30 times 53 head
↳ Not biased

$\boxed{p = 0.31}$ 32 times 60 head
↳ Biased

p-value is a measure of strength of evidence against the Null hypothesis that is provided by our sample data.

Interpreting p-value

↳ Reject H_0 when $\boxed{p \le \alpha}$

Without α

1) $p < 0.01$ → reject H_0
2) $0.01 < p < 0.05$ → moderate evidence
3) $0.05 \le p < 0.1$ → weak evidence
4) $p > 0.1$ → no evidence

p (0.01 to 0.05)

p-value in context of Z-test

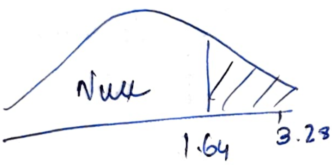
Rejection Region Approach :

$\mu = 50$ $\quad n = 30$ $\quad \bar{x} = 53$

$\sigma = 4$ $\quad \alpha = 0.05$

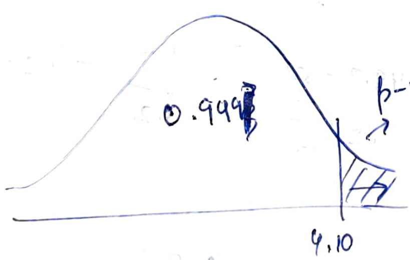
$\boxed{H_0 : \mu = 50 \quad H_a : \mu > 50}$

$$Z\text{-stat} = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} = \frac{53 - 50}{4/\sqrt{30}} = \cancel{3.28}\ 4.1$$



⇒ $\boxed{\text{reject } H_0}$

p-value Approach :



p-v 0.001 < 0.05

⇒ Reject my null hypothesis

Lays example

$\mu = 50$ $\quad n = 40$ $\quad \bar{x} = 49$

$\sigma = 5$ $\quad \alpha = 0.05$

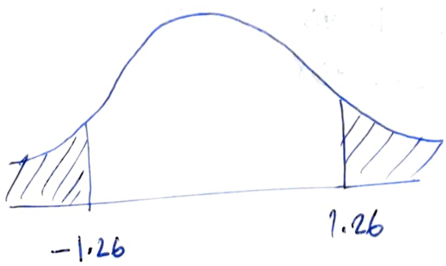
$\boxed{H_0 = 50 \quad H_a = \mu \ne 50}$

$$Z\text{-stat} = \frac{49 - 50}{5/\sqrt{40}} = -\frac{\sqrt{40}}{5} = -1.26$$

p-value = $a_1 + a_2$

$A(-1.26) = 0.103$

$A_1 + A_2 = 0.206 = p$



∴ $p > 0.05$

⇒ $\boxed{\text{Can't reject null hypothesis}}$

t-test — when we have sample std. deviation (s)

with p value — works good with small sample size

Three main types:

i) One-sample t-test → $H_0 \rightarrow \sigma \sim s$, $H_a \rightarrow \sigma \neq s$

ii) Independent two sample t-test → $H_0 \rightarrow S_1 = S_2$, $H_a \rightarrow S_1 \neq S_2$

iii) Paired t-test (dependent two-sample t-test) → $H_0 \rightarrow S_A = S_B$, $H_a \rightarrow S_A < S_B$

Single Sample t-test

Assumptions : 1) Normality
2) Independence
3) Random Sampling
4) Unknown population std

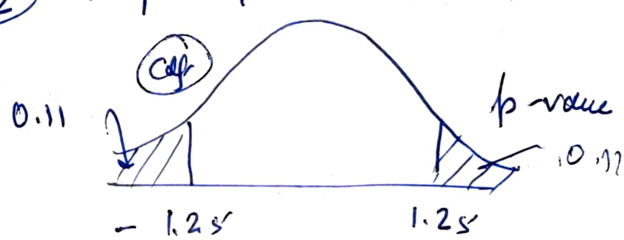
$\mu = 50$, $n = 25$, $\bar{x} = 49.7$, $\alpha = 0.05$

$s = 1.2$ → sample std

$H_0 : \mu = 50$

$H_a : \mu \neq 50$

Assuming it is normal



$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}} = \frac{49.7 - 50}{1.2/\sqrt{25}} = \frac{-1.5}{1.2} = -1.25$$

$df = n - 1 = 24$

$A = 0.11 \times 2 = 0.22 > 0.05$ (used cdf to calculate area)

$\therefore$ Can't reject H_0

Titanic Example

Shapiro-wilk test

→ 0.402

ttest_1samp, T-statistic = −2.22

p-value / 2 = 0.01768

Independent 2 samplest-test

Unpaired t-test, compare means of two independent groups

Assumptions for the test:

1. Independence of observations
2. Normality (Shapiro)
3. Equal variances (Homoscedasticity): F-test, levine test
4. Random Sampling

p-value < 0.05 ✗
p-value > 0.05 ✓

E.g., desktop users

$n_1 = 30$

$\mu_1 = 18.5$

$\sigma_1 = 3.5$ min

Mobile users

$n_2 = 30$

$\mu_2 = 14.3$ mins

$\sigma_2 = 2.7$ min

$H_0: \mu_d = \mu_M$

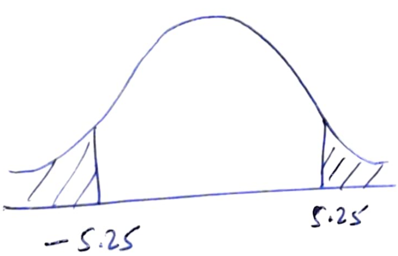
$H_a: \mu_d \neq \mu_m$

We will use significance level of $\alpha = 0.05$

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}} \quad \alpha$$

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{S_1^2}{\cancel{\sqrt{n}}\, n_1} + \frac{S_2^2}{n_2}}} = \frac{18.5 - 14.3}{\sqrt{\frac{3.5^2}{\cancel{\sqrt{}}30} + \frac{2.7^2}{\cancel{\sqrt{}}30}}}$$

$$= \frac{4.2}{0.80} = 5.25$$



$t = \cancel{t}\ -5.25$

$df = n_1 + n_2 - 2 = 60 - 2 = 58$

$p \sim 0$

Reject H_0

$H_0: \mu_m = \mu_f$

$H_a: \mu_m > \mu_f$ (Age)

$\alpha = 0.05$

Shapiro

p-value > 0.05 (Normal)

levine test

p-value > 0.05 (Equal variance)

Paired 2 sample - test (Dependent) → 1. Before - and - after studies
2. Matched or correlated groups

Assumptions 1. Paired observations

2. Normality : diff. should be approximately ~~norm~~ $D \sim N(\mu, \sigma)$

3. Independence of pairs : Each pair should be independent of other pairs.

H_0: No significant diff.
H_a: significant diff.

	before	after	diff
A	80	78	x_A
B			
⋮			
K	91	80	x_K

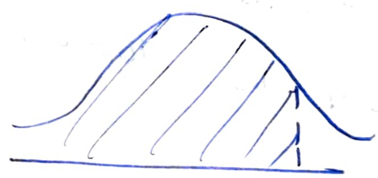
→ $\boxed{\text{(normal?)}}$

$\bar{X}_{diff}$: μ S

$$t = \frac{\bar{X}_{diff}}{S_{diff}/\sqrt{n}} = 0.104$$

p-value = 0.2 > 0.05

$\alpha = 0.05$



0.104

p-value = 0.54 > 0.05 → Null hypothesis can't be rejected